



Lecture 16

Models

Announcements

- **Project 1** and **Lab 5** due date **extended** to tomorrow 5pm
 - **HW 6** due Wednesday 2/28 @ 5pm
 - Midterm on Friday, 3/8, 7-9pm PT
 - Scope: everything up to and including lecture that day
-

Weekly Goals

- Wednesday
 - Methods of sampling
 - Distributions of large random samples
 - **Today**
 - Models that involve chance
 - Assessing the consistency of the data and the model
-

Review: Distributions

Distributions

- Any random quantity has a **probability distribution**:
 - All **possible** values it can take
 - The **probability** it takes each value
 - After repeated draws, it has an **empirical distribution**:
 - All **observed** values it took
 - The **proportion of times** it took each value
 - After **many independent draws**, the empirical distribution looks more and more like the probability distribution
-

Probability Distribution of a Statistic

- Values of a statistic vary because random samples vary
 - “Sampling distribution” or “probability distribution” of the statistic:
 - All possible values of the statistic,
 - and all the corresponding probabilities
 - Can be hard to calculate
 - Either have to do the math
 - Or have to generate all possible samples and calculate the statistic based on each sample
-

Empirical Distribution of a Statistic

- Empirical distribution of the statistic:
 - Based on simulated values of the statistic
 - Consists of all the observed values of the statistic,
 - and the proportion of times each value appeared
 - Good approximation to the probability distribution of the statistic
 - if the number of repetitions in the simulation is large
-

Assessing Models

Models

- A model is a set of assumptions about the data
 - In data science, many models involve assumptions about processes that involve randomness
 - “Chance models”
 - **Key question:** does the model fit the data?
-

Approach to Assessment

- If we can simulate data according to the assumptions of the model, we can learn what the model predicts.
 - We can then compare the predictions to the data that were observed.
 - If the data and the model's predictions are not consistent, that is evidence against the model.
-

Today's Examples

Some Goals of Data Science

- Understand the world better
- Help make the world better

For example

- Help expose injustice
- Help counter injustice

The skills that you have gained empower you to do this.

First Example

- U.S. Constitution grants equal protection under the law
- All defendants have the right to due process

We will study a U.S. Supreme Court case in the 1960s

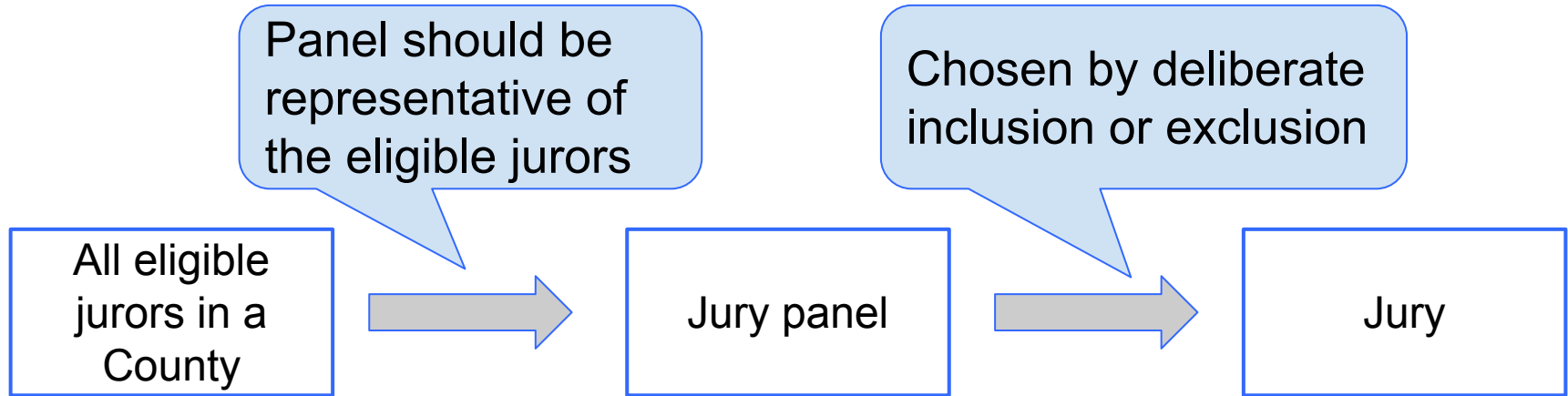
- A Black defendant was denied his Constitutional right to a fair jury
- The Court made incorrect and biased judgments about
 - the data in the case
 - the legal processes in the defendant's original trial
- We will discuss errors and racial bias in the Court's judgment

This case became the foundation of significant reform.

Jury Selection

US Constitution:

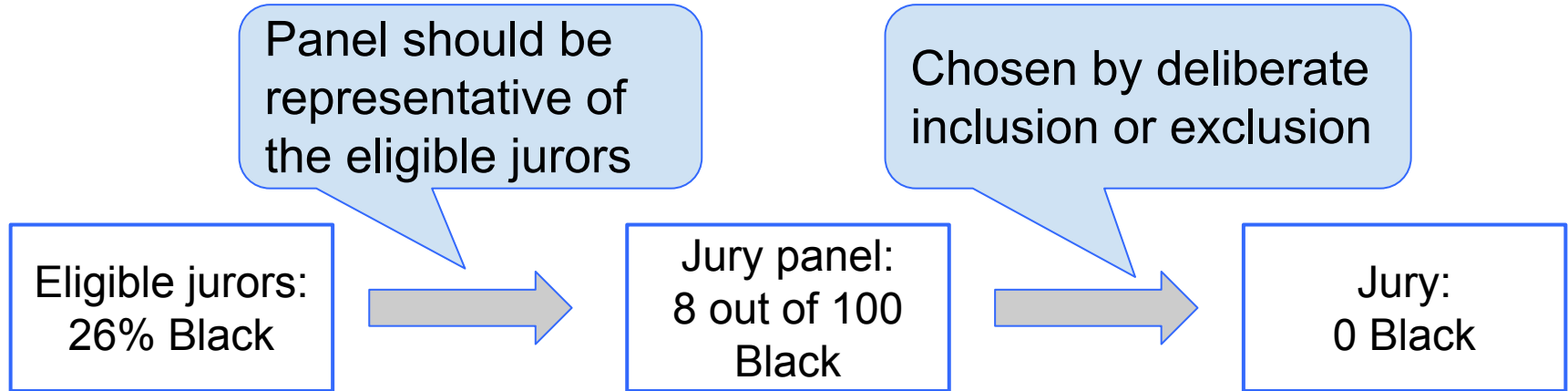
“right to a speedy and public trial, by an impartial jury”



Supreme Court Case

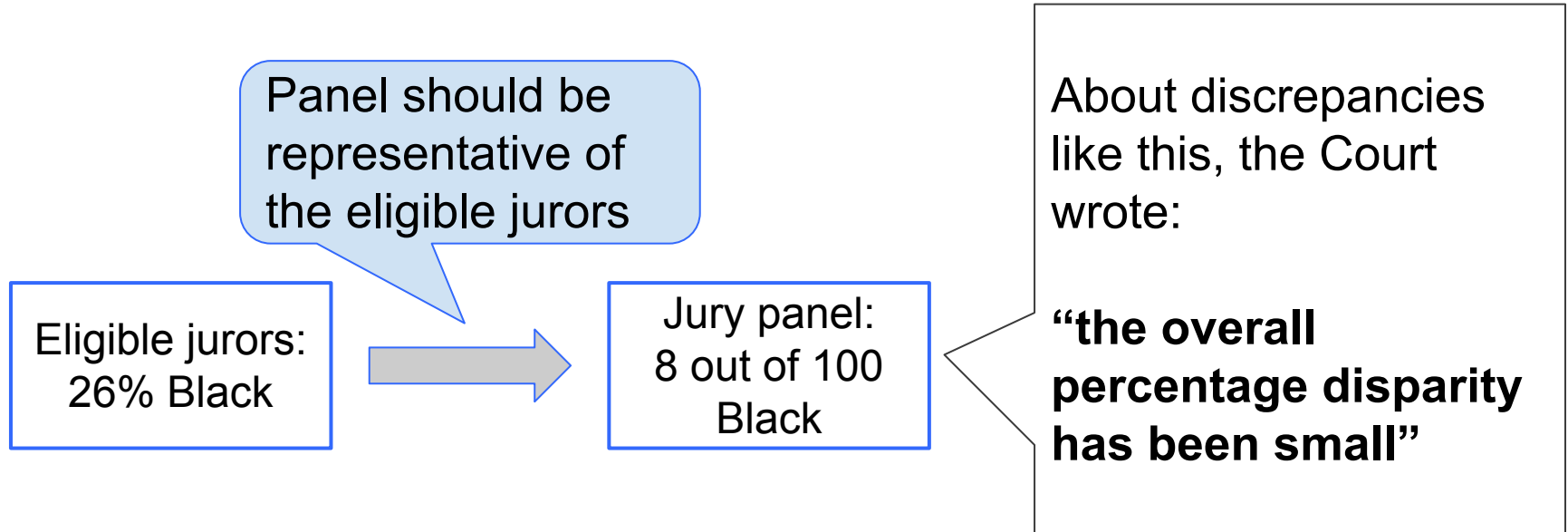
Robert Swain's Case

- Robert Swain, a Black man, was convicted in Talladega County, AL
- He appealed to the U.S. Supreme Court
- Main reason: Unfair jury selection in the County's trials



Supreme Court Ruling, 1965

- The Court denied Robert Swain's appeal.



Sampling from a Distribution

- Sample at random from a categorical distribution

`sample_proportions(sample_size, pop_distribution)`

- Samples at random from the population
 - Returns an array containing the empirical distribution of the categories in the sample

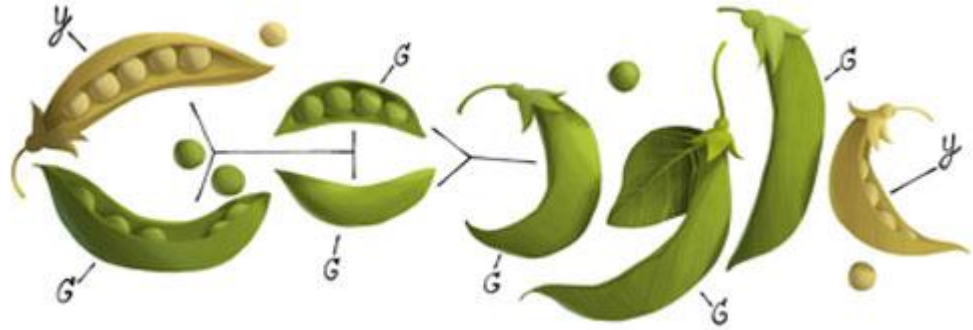
(Demo)

Statistical Bias

- Evidence provided by Robert Swain:
“only 10 to 15% of ... jury panels drawn from the jury box since 1953 have been [Black], there having been only one case in which the percentage was as high as 23%”
 - Percent of Black panelists was always lower than expected under random sampling
 - *Bias*: when errors are systematically in one direction
-

A Genetic Model

Gregor Mendel, 1822-1884



A Model

- Pea plants of a particular kind
 - Each one has either purple flowers or white flowers
 - Mendel's model:
 - Each plant is purple-flowering with chance 75%,
 - regardless of the colors of the other plants
 - Question:
 - Is the model good, or not?
-

Choosing a Statistic

- Take a sample, see what percent are purple-flowering
- If that percent is much larger or much smaller than 75, that is evidence against the model
- ***Distance*** from 75 is the key
- Statistic:
 - | sample percent of purple-flowering plants - 75 |
- If the statistic is large, that is evidence against the model

(Demo)

Two Viewpoints

Model and Alternative

- **Jury selection:**
 - **Model:** The people on the jury panels were selected at random from the eligible population
 - **Alternative viewpoint:** No, they were biased against black men
 - **Genetics:**
 - **Model:** Each plant has a 75% chance of having purple flowers
 - **Alternative viewpoint:** No, it doesn't
-

Steps in Assessing a Model

- **Choose a statistic** to measure discrepancy between model and data
 - **Simulate the statistic** under the model's assumptions
 - **Compare** the data to the model's predictions:
 - Draw a histogram of simulated values of the statistic
 - Compute the observed statistic from the real sample
 - If the observed statistic is far from the histogram, that is evidence against the model
-

Next time

RACIAL AND ETHNIC DISPARITIES IN ALAMEDA COUNTY JURY POOLS

A Report by the ACLU of Northern California

October 2010
